

MOVAID: a personal robot in everyday life of disabled and elderly people

Paolo Dario*, Eugenio Guglielmelli,
Cecilia Laschi and Giancarlo Teti
*ARTS Lab - Scuola Superiore Sant'Anna, Via
Carducci 40, 56127 Pisa, Italy*

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The growing number of disabled and elderly citizens, on one side, and the wide spreading of technology in everyday life, on the other, has led to a consistent effort devoted to the research of technological solutions for improving the quality of life of disabled and elderly people. Technology can actually provide a wide range of solutions, at different levels of complexity and cost. The recent progress of research in advanced robotics allows robotic solutions to be applied to assist disabled and elderly people in everyday life.

The MOVAID project, promoted by the European Commission within the TIDE programme, and co-ordinated by the Scuola Superiore Sant'Anna (Pisa, Italy), represents one of the first attempts to propose robotic assistance in such a personal sphere of activities as everyday life at home. Ten partners from five European countries joined the MOVAID Consortium, including Universities, validation centres, and industries.

The MOVAID project proposed, applied and validated some innovative concepts for the design and development of a modular robotic solution to the problem of personal assistance, by implementing a mobile robotic system and dedicated interfaces to standard appliances.

The final objective of the MOVAID project was to demonstrate two points. First, how mass consumer technological products, when accessible for disabled and elderly people, can enhance their level of autonomy in everyday activities. Second, how a robotic solution is not only technically feasible, but also acceptable from the user's point of view, if integrated in a modular assistance system.

The basic philosophy of the project relies on the concepts of 'design for all' and 'user oriented approach', as key factors for the introduction of technology in everyday activities. Such concepts were realised in the functional and physical distribution of the system in the house, including docking facilities for the mobile robotic unit.

The MOVAID system consists of a number of fixed workstations (PCs), located where main activities are carried out at home, such as the kitchen and the bedroom, along with a mobile robotic unit able to navigate in the house avoiding unexpected obstacles, to grasp and manipulate common objects and to dock to the fixed workstations for data exchange and power supply. Commands to the robot are given in a high level language through a graphical interface running on the fixed workstations. On the user interface, a continuous visual feed-back from on-board cameras is also shown to the user, allowing him/her not only to monitor what the robot is doing, but also to collaborate with it, by indicating objects and positions directly on the screen.

Moreover, to allow and ease access to standard technological products, interfaces for standard kitchen appliances were studied, and a prototype of a microwave oven interface, offering the oven basic functionality, was developed and tested.

Typical tasks for the system, defined on the basis of identified users needs, are: to warm up some food in a microwave oven and serve it at the user's bed; to clean the kitchen work surface; and to remove dirty sheets from a bed.

The developed prototype MOVAID system has been successfully validated with potential users in Italy, France and Switzerland, both through demonstrations and user trials, carried out in a residential house for disabled people in Italy.

The paper summarises the project and its achievements. The basic philosophy and the approach are introduced and a detailed description of the system is then provided, including the technical aspects related to the components design and development. Finally, the results of the validation phase on the system prototype are reported and discussed.

Keywords: Disabled and elderly people, personal assistance, robotic assistance, personal robot, design for all, user-oriented approach

1. Introduction

The application of technology to assist disabled and elderly people relies on two main factors: (a) the grow-

*Corresponding author: Tel.: +39 050 883400; Fax: +39 050 883402; E-mail: dario@arts.sssup.it.

ing number of people needing assistance in the industrialised countries [8], feeding the public awareness of the problem, and (b) the advance and spreading of technology in many aspects of the everyday life, which make a number of different solutions available even at low cost, and create the psychological substrate for favouring acceptability.

Even though long term solutions to the problems of disability should be searched within the frameworks of medicine and bioengineering, technology can provide short-term solution to improve quality of life.

The introduction of new technologies in the execution of everyday tasks presents some difficulties, especially in the activities carried out at home. Behaviour at home is highly subjective and hard to standardise. If users are disabled or elderly people, further difficulties are raised by the psychological implications of being 'forced' to rely on technology. Thus, satisfaction and enjoyment are important to consider, as part of optimising the level of interaction between the user and the product.

The progress of research in advanced robotics permits robotic solutions to assist disabled and elderly people in everyday life [1,7,14,17,25,27]. Research on rehabilitation robotics has achieved its most successful results in assisting people with disabilities in vocational environments and tasks. In particular, some examples of fixed robotic workstations have been realised, equipped with a robotic arm and peripheral devices [2, 13,15,26]. Typically, in these systems the workspace of the robot (mainly related to a worktable) is entirely structured, so that the robot operates in a repetitive fashion following pre-stored task trajectories. This spares the user from any kind of control effort. Systems for assisting the severely disabled to eat or perform common vocational operations (e.g., reading texts, using a computer or a telephone) have been implemented and widely tested with users [12,16,21,24].

Other solutions to the problem of assisting the disabled have been achieved with arm mounted wheelchairs. The usefulness of this solution relies on the user being able to move (or control the movements of) the wheelchair, so that the robot can be taken to the area(s) of activity. This solution became popular as it allows the user to operate the robot arm anywhere, rather than being limited to some fixed structured locations [18,23].

The most generally applicable solution is represented by autonomous mobile robots, since they can adapt to most personal and environmental situations. Their mobility and versatility make them particularly suitable

for severely disabled or bed-ridden users, as long as the interface between the user and the robot is easy to use. For example, the user should be able to train the robot with a high-level language, via a bi-directional user interface offering appropriate methods of input and suitable output.

Several prototypes and even commercial versions of mobile robots capable of performing a limited number of activities of daily living tasks have been developed. A sample is the HERO robot, realised by Heathkit, Zenith Corporation, for domestic and educational use. One of its most important characteristics is the ability to expand the limited basic capabilities through software and hardware enhancements. However, its commercial success was limited by very low performance compared to the expectations of the users.

Another interesting and more recent vehicle is the Labmate, developed by the Transition Research Corporation as a general purpose mobile base. It originated a successful commercial development, the Helpmate, for automatic transportation tasks inside hospitals and similar institutions. Helpmate is now operating in several American hospitals.

In the U.S.A., the Department of Veterans Affairs started a research program in rehabilitation robotics in the 1980's, aimed at improving the quality of life for disabled veterans. In this framework, the Veterans Administration Rehabilitation R & D Centre (Palo Alto, CA) developed the first prototype of the Mobile Vocational Assistant (MOVAR), which was the first mobile manipulator platform developed for rehabilitation application. Its potential value as a flexible tool for assisting the severely disabled has been deeply and positively evaluated with users [12].

The Italian project URMAD [3-5] led to the development of a prototype autonomous mobile robotic unit for assisting the disabled at home or in hospitals. The prototype, shown in Fig. 1, consists of a vehicle equipped with an eight degree of freedom robotic arm and a two degree of freedom gripper, a pan-tilt head supporting two television cameras and sensors for obstacle detection and localisation. Furthermore, a bed-side workstation allows the user to operate the robot via a man-machine interface based on a meta-language. Communication between the workstation and the mobile unit is achieved by a radio link.

This system was intended to be more a research platform than a product to be directly exploited in a commercial sense. Its main limitations are in its dimensions, weight and low autonomy due to high power consumption.

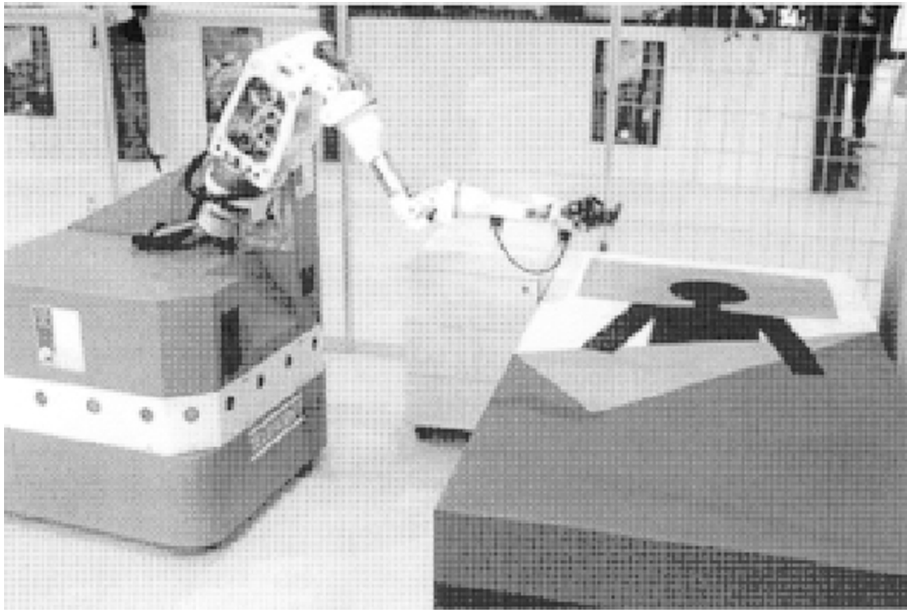


Fig. 1. The URMAD system while operating in a simulated household environment.

The MOVAID system concept combines the performance of a static robotic workstation with the versatility of a mobile, semi-autonomous robot. The system consists of static activity workstations, to which a comparatively simple mobile robot can dock. The mobile robot has the minimum technology required for it to navigate around the environment safely, but when docked, the combined robot and workstation offer the high performance of a static robot.

The basic philosophy of the MOVAID approach is described in the following Section. Section 3 describes the phase of analysis of user needs for MOVAID, while Section 4 gives a technical description of the MOVAID system. Finally, in Section 5 the validation phase is described and the results are reported and discussed.

2. The MOVAID philosophy and approach

MOVAID (MObility and actiVity AssIstance system for the Disabled) is a project partly funded by the European Union in the framework of the TIDE (Technology Initiative for Disabled and Elderly people) research programme from '94 to '97, for the development of a semi-autonomous modular robotic system for residential care of disabled and elderly people.

The basic philosophy of the project is based upon the concepts of 'design for all' and 'user oriented approach', as key factors for introducing technology in

everyday life. The project was more focused on the user's satisfaction with the interaction with the robotic device interaction, than on the efficiency and effectiveness of the device itself. Thus, user involvement is present at various stages of the project, to define the parameters of the product acceptability before its design and to validate its efficiency after the design.

The goals of the MOVAID project were achieved by taking two different (but related) approaches: first, the needs of severely disabled or bed-ridden users were satisfied by developing a robotic assistance system. Second, the majority of potential users were satisfied by the development of dedicated interfaces for consumer products.

In order to achieve its goals, the project has carried out a number of essential tasks, including: user requirement analysis and development of scenarios of use; definition of the MOVAID system based on user requirements and scenarios; definition of functional specifications; re-design of existing components; definition of design guidelines; prototyping of the MOVAID system; revision and improvement of systems based on user testing, followed by integration of systems; and user trials in Italy, France and Switzerland with elderly and disabled users.

In Fig. 2 the main phases of the project are shown, with particular reference to the phases of user involvement.

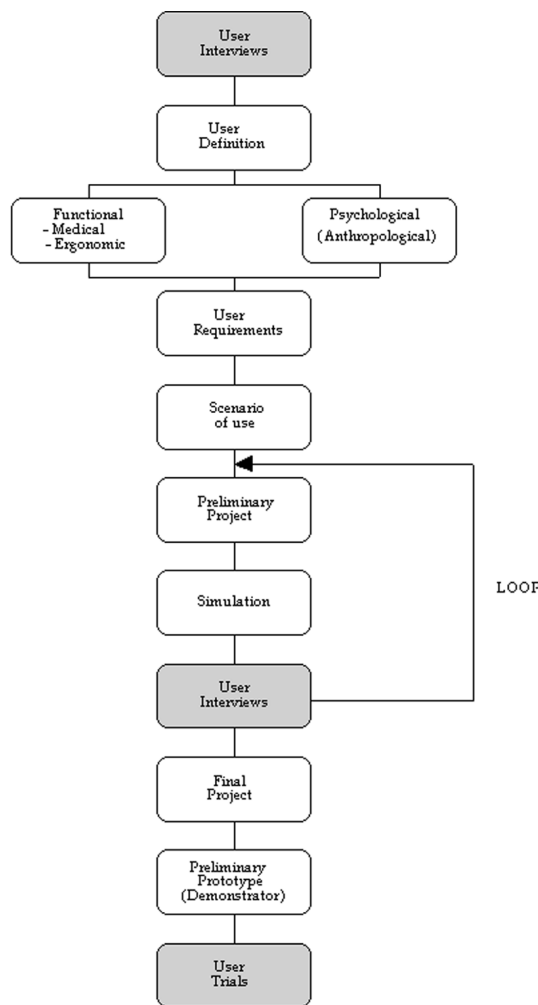


Fig. 2. The involvement of end-users in the various stages of the MOVAID project.

3. Analysis of user needs and design prescriptions

The analysis of user needs for MOVAID were carried out through interviews with disabled people, their relatives and assistants, in Italy, France and Switzerland [6].

The interviews were structured to be conducted by a research assistant, to take not longer than half an hour for each person, and to be completed, in each involved institution, within one week. A balanced number of males and females were involved.

The MOVAID interview questionnaire included questions on different areas, such as domestic environment, activity at home, aid and assistance received, technical aids, and personal ability. It also collected personal information such as age, gender and impairment.

There is an inherent difficulty in defining the acceptability of a technical device, or in accurately predicting the possible needs a new device can satisfy. Consequently, the first user interviews were intentionally designed with particular attention to the general approach and expectations of users towards technical aids, avoiding requests to define the precise tasks of the system. The questions were accompanied with iconographic material, to help determine the general attitudes about technical aids and robotics. For example, images of different robots (anthropomorphic, industrial, science fiction, toy) were presented, and questions about their general reactions (fear, funny, worry, interest, admiration) were asked. The results helped develop a profile of the user that is functional, as well as psychological and, in a wider sense, anthropological.

The survey results showed that most users feel positive about generic technical aids but consistently provide negative answers when robotic help is proposed. People seem to accept generic technical aids but dislike robot help. This could be due to limited exposure to negative information about robots, such as through science fiction or through industrial applications. In fact, the assistants confirmed that a deeper understanding of the meaning of “robot”, and an explanation of their real capacities made these attitudes more positive. Fig. 3 reports the most meaningful results from these surveys.

The results of the survey showed a wide range of possible users for the MOVAID system. The respondents present broad range of medical characteristics, age, cultural characteristic, and psychological attitudes. Instead of defining a precise, comprehensive, but enormously long list of individual user profiles, clusters of characteristics were grouped into some synthetic ‘user profiles’. The user profiles represent a design tool helpful in developing specifications for the system and as a reference for selecting users for the final validation phase.

A total of six user profiles were specified, three severely disabled and three moderately disabled. Each of them includes an indication of age and type of physical disability, intellectual attitude, degree of adaptation to their disability (related to the time or the level of disability), a dynamic parameter of adaptation, a psychological definition of the attitude towards disability, a dynamic parameter related to the need and possibility of learning with the system and, finally, an indication of the context of use (including helpers and the family as possible users, too).

An exhaustive list of the MOVAID system desired capabilities would be equally lengthy, so, tasks to be

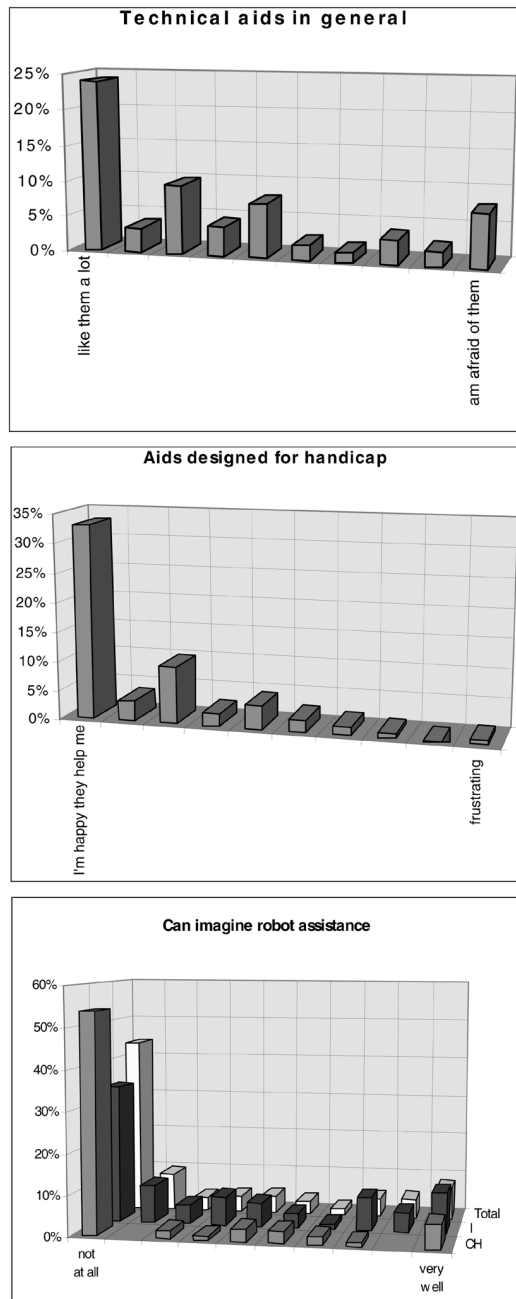


Fig. 3. The attitude of potential users towards technology.

performed by the system were grouped into three sample tasks. These three tasks address the daily activities of cooking, house cleaning and personal care:

1. warm-up a dish of pre-prepared food in the microwave oven and serve it at bed;
2. dispose of empty packages (or peels) on the kitchen work surface and clean it;

3. remove dirty upper and lower sheet from the bed and put them in a container.

Task 1 shows the system's ability to navigate throughout the house, including docking/undocking to fixed workstations, to grasp handles and objects, and to use a microwave oven. Task 2 mainly shows the system's ability to collaborate with humans and to recognise shapes. Task 3 shows how the system can perform tasks which are unpleasant for human carers, thus re-defining the helped/helper relationship.

The analysis of users' needs showed the importance of positive interaction between the user and the system. User involvement is highly desired in completing pleasant and satisfactory tests. However, system autonomy is needed in cases of emergency support for the user, or even for unpleasant, repetitive tasks. So, a qualitative scheme of user/system interaction was developed. The basic idea is 'modularity in autonomy', which permits the user to access the robot with different levels of involvement. Fig. 4 shows the different levels of user involvement/system autonomy related to different tasks. This concept was reflected in the design of the user interface, by developing four different levels of use, ranging from total autonomy for the system to total autonomy for the user. The four levels were named beginner, standard, advanced and expert.

Beginner level corresponds to maximum system autonomy. The user can only choose a pre-defined task to be executed. This level includes training procedures, self-explanatory commands, on-line help. Standard level is a good compromise of user involvement: the user still chooses a pre-defined task out of a list, but he/she can participate in the task if he/she wishes. For example, the robot may request user intervention to help localise objects or to recover from errors, although the robot has a default autonomous solution anyway. Advanced level allows the user to compose tasks by composing macros at a lower level, while user involvement is programmed in as a condition for the

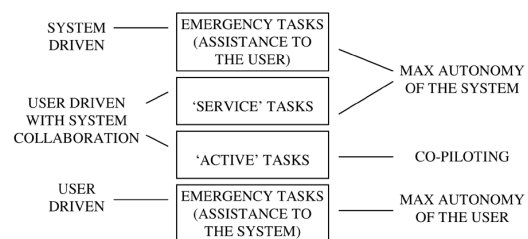


Fig. 4. Levels of user involvement/system autonomy in the execution of different tasks.

robot to accomplish tasks. At the expert level, the user completely controls the robot by composing tasks from single actions and names.

The screen layout was designed to be as user-friendly as possible, and to not look too 'computer-like'. A big window contains the image acquired by the on-board camera, so that the user can see what the robot is seeing, and can interact in the robot workspace through the shared image.

4. The MOVAID system

The MOVAID system is composed of a distributed robotic system, including a mobile robotic unit and a number of fixed workstations, and of dedicated interfaces for standard kitchen appliances. This section provides a technical description of the MOVAID system components. We describe the mobile unit, the fixed workstations and the dedicated interfaces, separately.

Some of the MOVAID components that already existed (i.e., the mobile base, the robotic arm and the hand) were re-designed to improve aesthetic appeal.

4.1. Mobile unit

The MOVAID mobile unit is composed of a four-wheel mobile base supporting a robotic arm and hand. Mounted on the first link of the arm are a pan-tilt head supporting two television cameras, and a Distributed Local Positioning System (DLPS)-based self-localisation system. The mobile base is also equipped with a ring of ultrasound sensors for obstacle detection, which is partially covered by a tray for object transportation. An active bumper around the vehicle halts the system in case of collisions. The male docking system is mounted on the rear lower part of the mobile base. The mobile unit structure is modular, so that the vehicle can be used on its own, as a mobile platform, or the arm can be mounted on a different support (a table or a wheelchair).

The vehicle used as a mobile platform for the MOVAID system is the Electric Pedestrian (by INSERM - Unité 103, Montpellier, France). The Electric Pedestrian, re-designed to give it a warmer and more pleasant look, is a four-wheel vehicle with two steering front wheels and two driving rear wheels. The electronics for driving the vehicle are located in the lower part, while the main compartment contains the batteries in the front, and the computing unit of the mobile unit (i.e., two PC 104 racks) in the rear.

The MOVAID arm is the DEXTER arm derived from the prototype developed by Scienza Macchinale s.r.l. (Pisa, Italy) for the URMAD system. It is an eight degree of freedom robotic arm especially aimed at operating in unstructured environments. The redundancy in its kinematic structure facilitates dextrous manipulation, enables the arm to be configured for work at various heights above the ground. It also allows the arm to fold to minimise its work envelope. The physical structure of the arm replicates the anthropomorphic structure of a trunk and an arm with shoulder, elbow and wrist. The eight motors are all on the first link, where the weight is concentrated, with transmission achieved through cables.

The arm end effector is the three-fingered gripper developed by Scienza Macchinale s.r.l. (Pisa, Italy) in the MARCUS project. The Marcus Hand is a two-degree of freedom prosthetic hand possessing great gripping capability in terms of set of graspable objects, size and grip force. A passive mechanical compensator allows to fit the prosthesis kinematic configuration to the object, and the reduction gears of the thumb and fingers are equipped with an irreversibility mechanism that makes possible to maintain the grip force even though the motors are powered off. Force sensors are mounted on the three fingertips in order to control the force of the grip.

The MOVAID head, developed at the University of Genoa, is a pan-tilt device (two degrees of freedom) supporting a binocular TV camera system. The head is located on the top of the first joint of the arm and includes a small platform for supporting the DLPS localisation system.

The DLPS localisation system developed at the University of Genoa allows the mobile base to compute its absolute position and orientation in the environment. It is an active positioning system for indoor applications in multi-robot environment, using a modulated light beam. The system is composed of two parts: an on board rotating unit and a set of active beacons distributed over the operating area (located on the walls at suitable heights). The rotating unit transmits a rotating and modulated beam of light (red laser) and it has an omnidirectional infrared receiver. The active beacons are little boxes that receive the beam of light transmitted by the rotating unit and reply with their identification codes. The reply messages from each beacons are processed to reconstruct the position and orientation of the robot within the space.

The mobile robotic unit can dock to a fixed workstation through a purposively designed docking system,

composed of a special connector for transmission of power and data, and a two degree of freedom mechanical support. This allows connection even at low positioning accuracy. The docking system has been designed so that female part affixed to the workstation occupies little space. The docking system is also used to correct for any inaccuracies within the absolute localisation system on board the robot. These inaccuracies are due to errors in the position of the beacons, low accuracy of the rotating part, and low accuracy of the odometry.

An Ethernet radio link allows bi-directional data flow between the fixed workstations and the mobile base when the robot is undocked. The fixed workstations send commands to be executed by the mobile base through the radio link, while the mobile base sends back the result of the command executed and the images acquired by the television cameras.

The hardware architecture of the mobile unit is composed of two PC 104 racks, connected through a serial link. The on-board resources are distributed onto the two racks, one of which communicates with the fixed workstations through the Ethernet link (either radio or physical, through the docking connector). The first rack contains the electronics for the base, hand and US control. The second rack, the one with the radio link, contains the hardware for the control of the arm and the head systems, and for image acquisition. Some photographs of the MOVAID mobile unit, in the demonstration environment where it has been validated, are shown in Fig. 5.

4.2. Fixed workstations

Two fixed workstations are located within the household demonstration site: the main workstation is located in the bedroom, close to the bed-ridden user, while the second workstation is located in the kitchen [19].

Each fixed workstation contains a PC, which houses the software modules for the system that are not required to be on-board. The two fixed workstations are connected into a local network by an Ethernet link, and are connected to the mobile base by an Ethernet radio link. The female sections of the docking system are located near the fixed workstations, one on the bottom of the kitchen furniture and one under the bed.

The fixed workstations permit the user to access the robot through a graphical man-machine interface, which can be used either by means of the standard PC mouse or by a specific human-machine interface. In

fact, an M3S keyboard and mouse emulator allows access to the man-machine interface by means of any M3S input device. The M3S standard, developed through an EU funded project, allows any M3S output device to be controlled by any M3S input device. The M3S concept is based on the need, expressed by people with disabilities, to own and use one input device that is suited for their functional abilities and personal preferences. Being able to control different devices through one input device avoids the difficulty of acquiring, learning and using new input devices for each possible new assistive device [22]. The MOVAID user can use his/her own M3S input device for accessing the MOVAID user interface, through the M3S mouse and keyboard emulator. Fig. 6 (a) illustrates the scheme of the M3S system while Fig. 6 (b) shows a photograph of the prototype of the M3S mouse emulator.

The instructions given by the user are processed by the proper software modules, implemented on the fixed workstations, and then translated and transmitted in the format required by the low level controllers on board the system [11].

The Multimedial Man-Machine Interface (MMMI) allows the user to access the MOVAID system, providing tools for requiring tasks. Moreover, the MMMI provides the user with information on the system status. Different levels of use were previously defined to better satisfy user needs. The lower levels of use are characterised by high autonomy for the system and little or no involvement of the user during the execution of tasks. The lower levels are most appropriate for new users or for users who don't want to be involved in the execution of tasks. The higher the level of use, the less autonomy for the system and the more involvement of the user.

Besides providing a user-system interaction tool, the MMMI also provides a continuous flow of visual data, coming from the on-board cameras. Such information is useful both for allowing the user to monitor the robot behaviour, and, mainly, for allowing him/her to help the robot during the execution of tasks. The idea is that the user, seeing what the robot is seeing, could ask for tasks involving observed objects and could help the robot identify objects by indicating them directly on the image he receives as visual feed-back via the MMMI. Pictures of the MMMI screen layout, at different levels, are given in Fig. 7.

The Supervisory Modules (SM) on the activity workstations plan, schedule and control the execution of all the tasks performed by the system, manage errors and monitor critical situations. The supervisory modules



Fig. 5. The MOVAID prototype performing the typical tasks in the demonstrative environment.

directly communicate with the MMMI, by means of a special synthetic language. The MMMI gives a preliminary interpretation of the input from the user, by translating it into the synthetic language, and transmits user requests to the supervisory level. When a task is requested by the MMMI (expressed in the synthetic lan-

guage) the supervisory modules interpret this request and produce a sequence of commands for the suitable low level functional module.

The Low Level Functional Modules (LLFM) are the modules allowed to access to the different MOVAID effectors, by invoking a set of Low Level Functions.

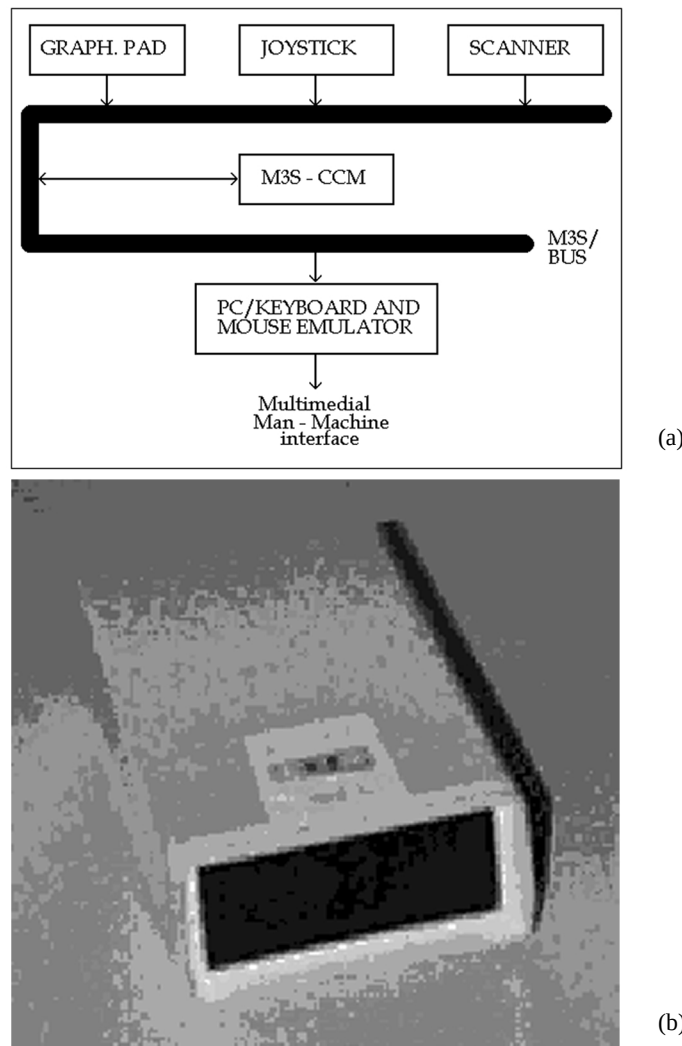


Fig. 6. (a) The M3S configuration of the MOVAID system and (b) a photograph of the emulator prototype (by FST, Neuchatel, Switzerland).

They translate the Low Level Commands received from the Supervisory Modules into instructions to the Low Level Functions, in the proper syntax and parameters.

The high level Communication Manager module (CM) is responsible for managing communications among the fixed workstation and the low level software modules, by allowing data to be transmitted through the Ethernet Connection, from the workstation core to each low level end-effectors controllers.

Finally, the system Data Base (DB) contains the data that are shared among different modules.

4.3. Dedicated interfaces

As for dedicated interfaces to standard appliances, the MOVAID project concentrated on two products:

a microwave oven and an electric citrus press. Each product required different system requirements [10].

With an appropriate interface, the microwave is a useful tool for moderately disabled and elderly people, offering them increased independence in an important everyday activity. Potential benefits to users are both functional (i.e., able to cook) and psychological (e.g., feelings of independence and achievement; able to cook for others).

The citrus press offers the 'luxury' of fresh juice, but its interface denies many moderately disabled and elderly people access to both its 'rational' function (i.e., the production of fresh juice), and its emotional function (i.e., expressions of health, and 'caring' through making juice for others). Suitable assistive robotic de-

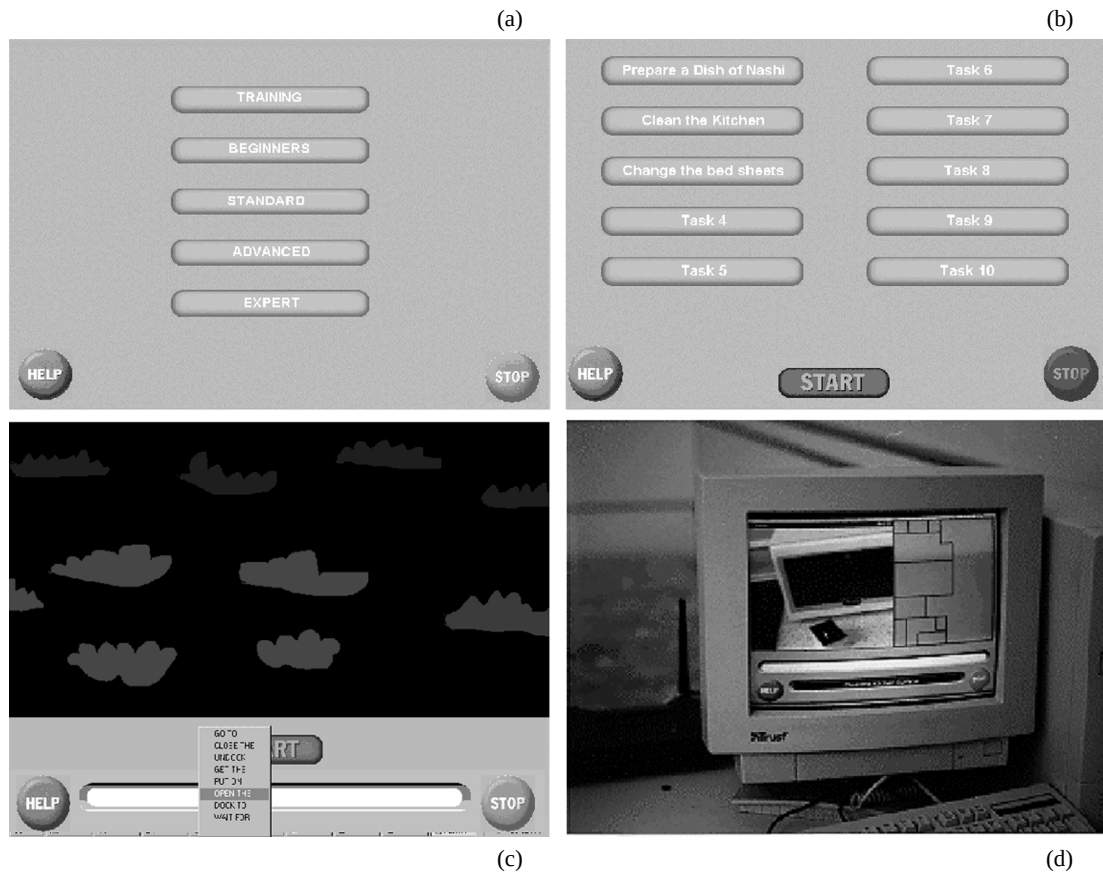


Fig. 7. Some examples of the MMMI screen layout. (a) Selection of user level; (b) selection of a task at beginner and standard level; (c) composition of a task at advanced level; (d) screen layout during the execution of a task.

vices would enable disabled and elderly people to enjoy these functions and their associated benefits.

The problems with standard microwave ovens for the MOVAID project centre on the user interface. Although only the power and the heating time need to be set, interaction with the oven is perceived as more 'technical' than with a conventional cooker. Reasons for this perception include: most ovens look more like a TV than a cooker, controls and displays are normally located on the front of the oven, control panels are 'mysterious' and not supportive. Furthermore, people with physical or cognitive disabilities are not considered in the design of controllers, as most controls require fine motor control and good vision for their operation. The sequencing and layout of controls are also poor, presenting problems for all users. In summary, the interface for microwave ovens expresses the 'engineer's viewpoint', in the sense that the user is offered controls which operate variables. This is fine if the user understands the variables, and what they are for, but it is

an industrial approach, which is not really appropriate for domestic users.

The problems with the citrus press considered in the MOVAID project, concern cutting the fruit safely, picking up the cut fruit, pressing the cut fruit to the cone with minimum effort, and then disposing of the pressed fruit rind.

A study on the different sensorial and/or motor impairments of users has been the baseline for the design of the appliance interfaces, including a study on the possible scenarios of use. Fig. 8 (a) summarises the tools required by the various users of the citrus press, while Fig. 8 (b) summarises the different modules required by the various users of the microwave oven.

The resulting MOVAID microwave oven interface, which is shown in Fig. 9 (a), is composed of 3 functional elements: the power control, the time control, and the pre-sets. The power control is in the form of an Archimedean spiral, conveying increments of heating power both physically and visually, and a 'flame'

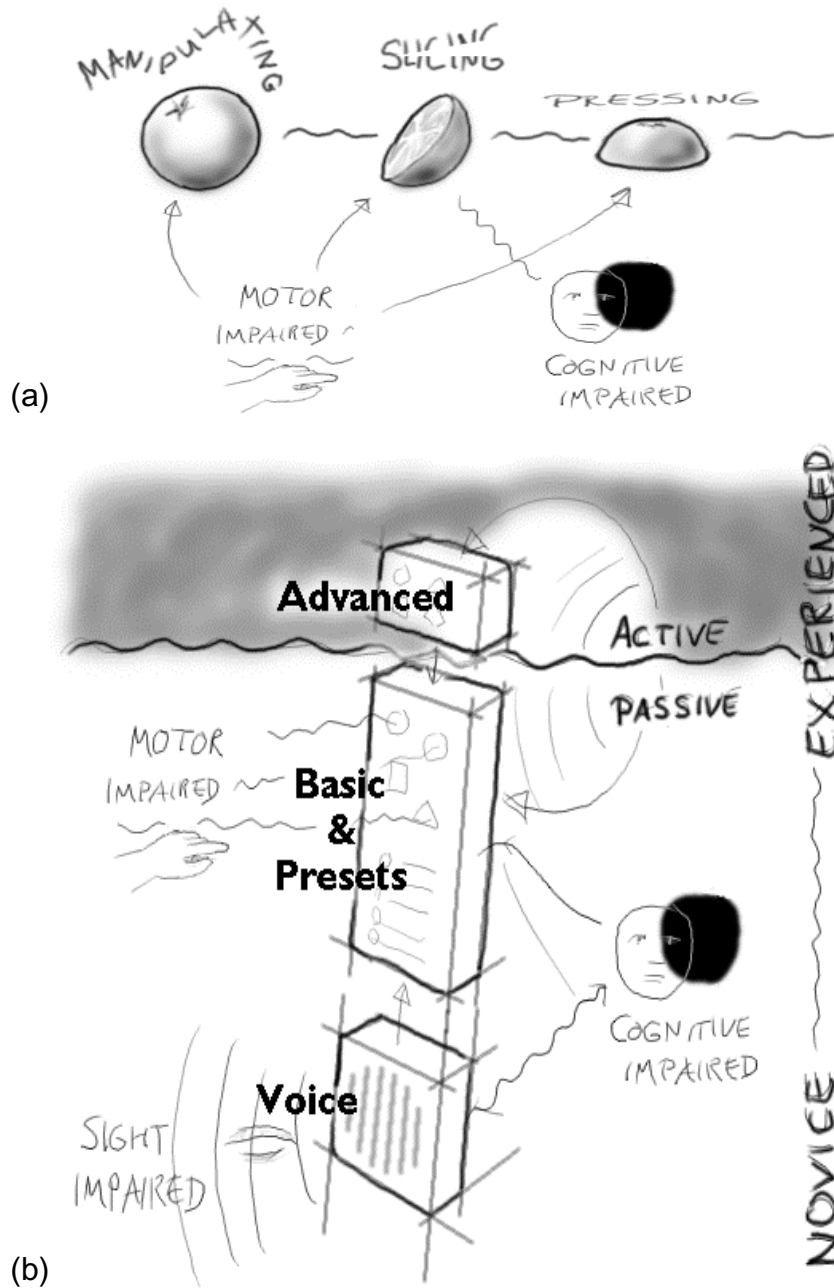


Fig. 8. (a) The concepts of the tools required for the citrus press and (b) the functional modules for the microwave oven interface (by Philips CD, The Netherlands).

formed lever rotating around this as a recognisable 'power' symbol. The time control resembles a clock-face with an external hand. The glass face allows a graphical representation of cooking time and a series of studs around the edge of the face provide a tactile feed-back. The hand sets the time by rotating it around

the face, and starts the cooking process by tapping it in the direction of countdown. The pre-sets were produced as a key system: special keys with a hole code are inserted in the keyhole in the body of the unit.

The tools for the citrus press consist of a universal holder, a cutting board and a pressing aid (Fig. 9 (b) and

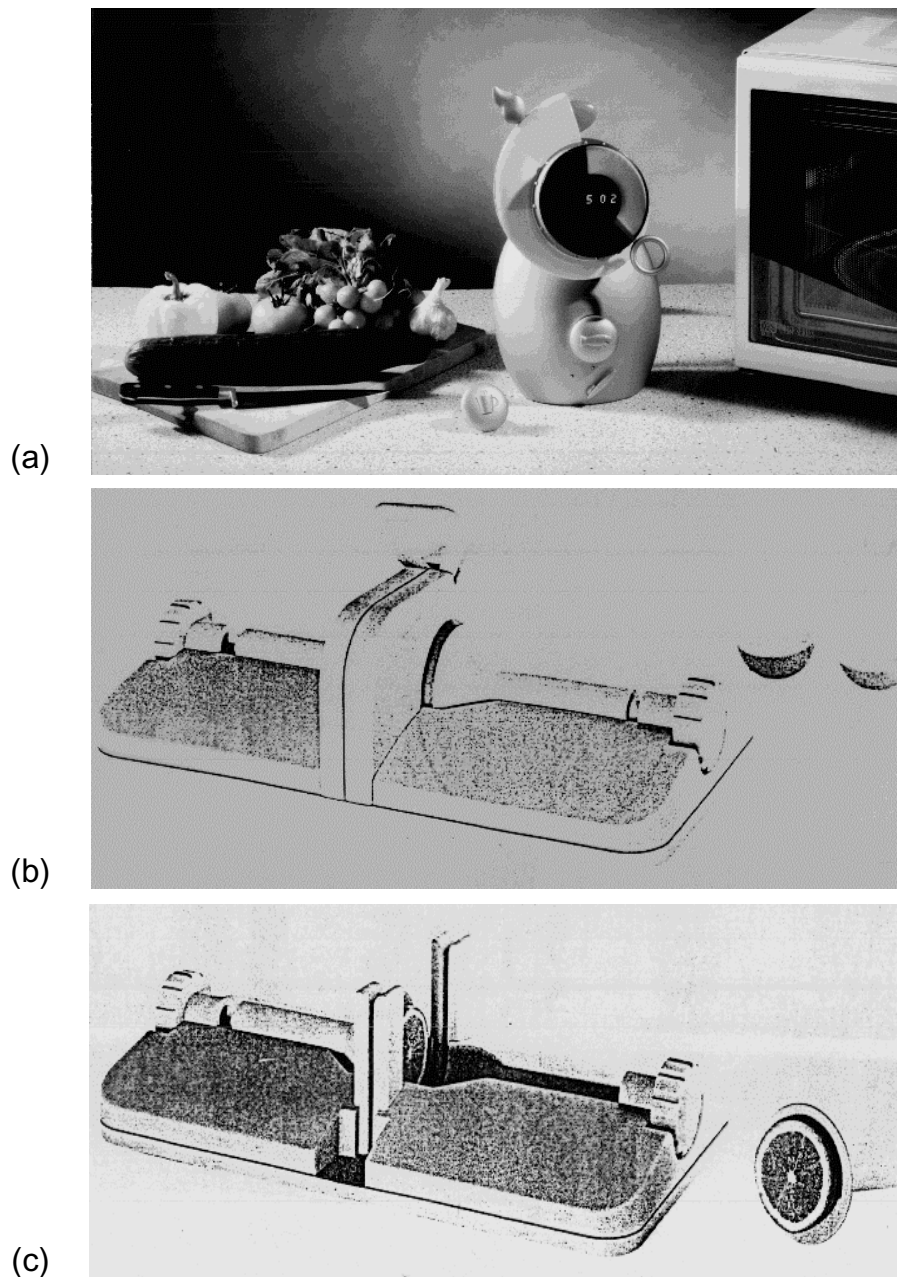


Fig. 9. (a) The MOVAID microwave oven interface and (b) the MOVAID tools for the citrus press: the universal holder in the cutting board and (c) holding the cut fruit.

(c)). The universal holder holds the fruit, allows the user to manipulate it, and then transports and presses it. The lower part is composed of a cone containing pre-positioned retractable barbed pins that pierce and hold the fruit. The universal holder places the fruit directly on the cutting board. The cutting board has another similar but opposite inclined holder which also holds the fruit, so that it can be easily cut by hand or by the

incorporated knife. The universal holder then presses the fruit against the citrus press cone.

5. Validation of the MOVAID system

A validation phase was planned for the final period of the project to evaluate the MOVAID system through

user trials with disabled and elderly people, in Italy, Switzerland and France. The six sample user profiles and the three typical tasks resulting from the analysis of user needs provided the basic references for the evaluation.

In Switzerland the interface for the microwave oven was evaluated with a group of 13 elderly people [9]; in Italy the whole system was evaluated with a total of 35 disabled people and assistants [20]; in France the system was evaluated through photographs, a video and a simplified version of the MMMI, with a wider group of 16 people with spinal cord injuries [20].

5.1. Validation in Switzerland

The evaluation of the microwave oven interface by elderly people had three objectives:

First, determine to what extent elderly people need to use a microwave oven, what problems they meet when using it, what assistance MOVAID provides in addressing their specific problems. Second, determine how elderly people perceive microwave ovens, what advantages and/or disadvantages the dedicated interface presents in comparison with the traditional microwave oven, to what extent it meets the specific needs of the elderly, and what changes should be made to satisfy their requirements. Third, ensure that the interface instructions are appropriate, easily understood by elderly people, and fulfil their need for information and feedback.

5.1.1. Methodology

The evaluation of the MOVAID dedicated interface involved two stages of interviews with various people concerned with the relationship between the elderly and microwave ovens, or with the MOVAID dedicated interface. Participants represented four groups: elderly people not owning a microwave oven; elderly people already owning a microwave oven; home care professionals; appliance sellers. We also conducted telephone interviews with two home delivery meal companies, and conducted personal interviews with the directors of two nursing homes.

Stage one interviews developed general information about how older people used microwave ovens as a tool without trying to be exhaustive. Stage two collected the specific opinions of elderly participants about the MOVAID dedicated interface, with some respondents having a microwave oven and others not. The questions explored the ergonomics of the device, the personal adjustments to problems that people make as they lose their autonomy, changes to the system design or to the user instructions.

5.1.2. Results of evaluation of the microwave oven interface

The stage one interviews revealed that very few elderly people have a microwave oven, at least in the urban environment where the interviews occurred. Summarising the results, microwave ovens correspond to the life style of active people looking for efficiency, which is not the case of elderly people who rather have too much time on their hands and only a few activities to complete. Elderly people who are losing their autonomy face many difficulties in learning new tasks. Microwave ovens also introduce a change in their food culture such as methods for cooking and the type of food prepared. Elderly people are cautious about change and microwave ovens represent a technical modernity. Nevertheless, elderly people are conscious of the advantages of microwave ovens. Those who do not have one plan to buy one when they have to, with the technical and sometimes financial help of their relatives. Home care professionals consider microwave ovens to be a viable solution to the cooking problems of elderly people losing their ability to cook independently.

Summarising the results of the stage two interviews, we should note that people already equipped with a microwave oven, home care professionals and electronic household appliance sellers find the MOVAID dedicated user interface attractive when it is sold at the same price as a traditional microwave oven.

The advantages of this system are numerous: simple access, clear controls, easy key system, ergonomics, large controls, separated interface, security, and clear instructions. People liked the design even though they found it unusual. They made several suggestions to improve the interface, such as integrate controls, reduce its size, add a power indicator, change colours, and design enhancements.

The results of the stage one and stage two interviews were analysed to validate the original aims of the MOVAID project. The main aim of the MOVAID project (i.e., providing users a more comfortable and enjoyable access to standard consumer product), was achieved by the microwave oven interface, which made the microwave oven easier to use for elderly people, people with physical or cognitive impairments, and also for children. The controls were found to be simpler than the controls of a traditional microwave oven, more appropriate to wavering and inaccurate movements, more ergonomic and more open to operation by “intuitive” logic.

In summary, the evaluation with elderly people showed that they do not typically use microwave ovens;

the problems with traditional microwave ovens are mostly related to the user's lifestyle, and the complexity of the appliances available on the market. The MOVAID system is most useful for elderly people who are no longer able to cook themselves, who receive home delivered meals, and/or who heat up meals prepared by someone else.

As far as the dedicated interface/microwave oven group, the system is much simpler and ergonomic than traditional microwave ovens, it allows an intuitive use, and it comes with useful didactic instructions. However, the device appears to disturb the technical and aesthetic environment of elderly people, because of its size, the multiple components and lack of specific user features.

The different categories of elderly people (equipped and not equipped with microwave ovens) did not agree about the ideal microwave oven control. Their observations allowed us to identify possible future improvements of the MOVAID's dedicated interface:

- the controls should be as simple as the MMI but integrated in the microwave oven's body;
- the colours symbolising power level and time settings should be selected according to their social meaning (e.g., cold = blue and hot = bright red, with the remaining time indicator in another colour);
- the key control of the microwave oven should be integrated in the body of the microwave oven, so the selected key could be introduced directly in the device;
- the microwave oven should have a food weight factor;
- the price of the group should be the same as a current device with similar functions;
- the device should come with simple and didactic instructions.

5.2. Validation in Italy and in France

Validation in Italy was carried out at O.A.M.I. 'Casa Angela', in Livorno, where the MOVAID system has been installed, in collaboration with the Local Health Care Management Unit (ASL 6, Livorno) which provided three therapists for the definition of the validation protocols, the selection of users and the collection and analysis of results.

The main objective of the Italian validation was the evaluation of a number of features of the system, such as:

- accessibility,
- acceptability,
- helpfulness,
- functionality,
- friendliness,
- easiness of use,
- easiness of learning the use,
- easiness of interaction.

The validation was carried on along two stages: the first stage consisted of trials on the MMMI as a stand-alone device, while the second stage consisted of trials on the whole system. Two different evaluation forms and one questionnaire were prepared for the two stages of validation: one evaluation form for stage one and one evaluation form and the questionnaire for stage two.

The forms allowed the evaluation to occur by direct observation, and with the forms completed during the trials by the therapists or the technicians involved. The questionnaire allowed the evaluation to record the user's opinion by having them completed right after the trial. A special section was included in the questionnaire to capture the family member/assistant opinion, to evaluate the observed benefits of system use.

In stage one, the user received a brief demonstration and explanation of the functionality of the MMMI and was then asked to complete one of the three typical tasks, at each level. The observing therapist completed the proper evaluation form during the trials.

In stage two, the users received a demonstration of the system by means of a video, with explanations on the device overall performance from a functional point of view. Then the users were presented the integrated system in the demonstrative environment. Users were asked to request that the robot complete one of the three tasks, and to collaborate with the robot when required. In stage two, an evaluation form was filled out by observing therapists or technicians during the trials, and a questionnaire was filled out by the user and possibly by his/her personal assistant, after the trials.

The selection of participants for the Italian validation was based on the three severely disabled sample profiles. One preliminary result was that nobody corresponding to one of the profiles, including a temporary disability, agreed to participate in the validation. As a consequence, the user group was enhanced with people beyond the user profiles to include a wide range of disabilities, ages, and psychological attitude towards technology. People who had never used a personal computer before were also included.

The validation session in France had the same objectives of the Italian validation, but it was carried out

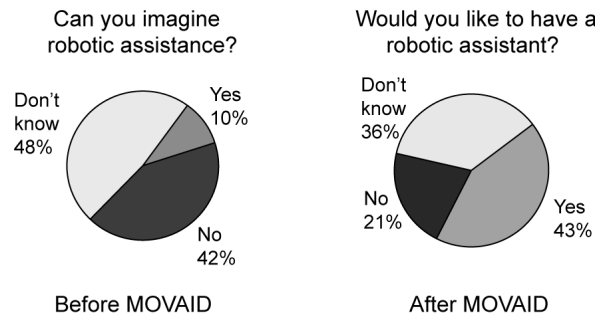


Fig. 10. Different attitude towards a robotic assistance before and after the MOVAID experience.

with a different methodology and on a different user group. Due to logistic constraints, the MOVAID system could not be moved to France, so the French evaluation was carried out with the support of illustrative material (such as photographs and a video); while the validation of the MMMI was carried out on a simplified version of the MMMI, run as a stand-alone application. The employment of such demonstrative material, and the availability of French rehabilitation centres, allowed to increase the number of people involved.

As a general result, the system was enthusiastically accepted especially by users with severe motor impairments and by those with poor or no upper limb function. The system was judged as a helpful, but not strictly needed tool by paraplegic people with full ability of upper limbs. The visual feed-back on the screen was confirmed as a comfortable feature because it allowed the user to monitor the robot, and the point-and-click mechanism of interaction on the image was appreciated because it did not introduce additional burdens to the user's standard activity.

When searching for common features in the results across the different evaluation sessions, we observed that the level of user interest was quite different between the first analysis of user needs (see Fig. 10) and the later validation phase. In the early analysis, we presented something to users which was almost completely unknown. This created a level of mistrust in the system. In the validation phase, users were asked to evaluate something much more concrete, which they could see while operating (live or in a video). The opinions of users, in the validation phase were much more positive. Also the elderly people participating in the Swiss validation showed more positive interest when they already knew and used microwave ovens, while negative opinions came from those who never used a microwave oven before. Further, the experience motivated most users to propose new tasks, and suggest

modifications and improvements related to their own needs.

Nevertheless, some users gave negative opinions on the MOVAID system, mainly related to a general mistrust in the real technical performance, but also for a sort of 'fear' for depersonalisation of their personal assistance. Also, some French users expressed their preference to focus research on medical solutions to the disability itself.

The aesthetic design of the system stimulated many comments and suggestions, but it seemed to achieve the proposed goals of giving the system friendliness and an appliance-like appearance. This appearance increased its acceptance for use within a domestic environment.

6. Conclusions

This paper presented the TIDE - MOVAD project as the European attempt to propose robotic solutions to the problem of personal assistance to disabled and elderly people.

The real challenge of the MOVAID project was to demonstrate that a robotic solution is feasible, not only from a technical point of view, but especially as a product that can improve the quality of life for disabled people and be accepted by the final users.

Potential final users were the focus of the project and a number of disabled and elderly people were involved in the project activities since the very beginning. A total of 207 final users from three different European countries participated in the project in total, including the user need analysis phase and the validation phase. One of the most unique aspects of the MOVAID project was the presence of industrial designers in the project, who helped us maintain a strong link with the final users and to mediate the technical activity with the users' needs.

The validation phase clearly demonstrated the validity of the MOVAID concepts and gave important indications for the pursuit of activities in the field of Rehabilitation Technology in general and Rehabilitation Robotics in particular. The MOVAID project indicates that a complete solution to the requirements of the moderately and the severely disabled for daily living and vocational task performance, requires a combination of existing products, including telematic, domotic¹ and robotic solutions. Further, to be effectively operated by the end user, these solutions must work in co-operation through a standardised command protocol and user interface.

From a technical point of view, refinements to MOVAID will aim to increase performances and to reduce costs, by relying on commercial components. We will continue to emphasise extended validation with end users, to maintain our user oriented approach and staying current with user needs as they evolve and change over time.

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¹Domotics is a newly-born word (from the French Domotique) widely used in Europe for indicating the automation and control of the functions of the house.

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